

MPX NYC Appendices

(A) Introducing SSNAC

a. Outline

b. Problem: The role of context (shared)

- no context
- social context
- ecological context
- spatial context
- social, ecological, spatial context.

c. Next up

(B) Person-place Graph

a. Outline

b. Problem: context is complicated, data are complex

- current practice is tables
 - different tables for different types of entities
 - expensive joins and lookup
- graph simplifies and makes efficient
 - data object: graph
 - no joins, network-based lookup
 - > example graph database

c. MPX NYC person-place graph

- node table
- edge table
- related graph: person
- related graph: place

d. Next up

(C) Network measures

- a. Outline
- b. Problem: relationality
 - spillover: infectious disease
 - interference: mental health
 - zero-sum games: racism
- c. Place and person measures
 - place social reach
 - place spatial reach
 - person social reach
 - person spatial reach
- d. Graph (system-wide) measures
 - spatial mixing coefficient
 - proportion in CC
- e. Next up

(D) Causality

- a. Outline
- b. Problem:
 - SUTVA
 - Extended Dag
- c. Classic model
 - Independence assumption
 - Identical distribution assumption
 - Mechanistic assumptions
- d.
 - implications for analysisNetwork model
 - network conditional independence assp.
 - network homogeneity assumption
 - mechanistic assumptions
 - implications for analysis
- e. Graph-level interventions and outcomes
 - mpx nxc hypothetical intervention
 - causal DAGs
 - Causal estimands
 - coverage
 - connectedness
- f. Issues in estimation
- e. Next up

- Network Measurement ~~and mutation~~
- (E)
- outline
 - problem: no network data
 - MPX NYC spatial network mapper
 - family networks
 - spatial data
 - passive data
 - sp next

(F) Implications

(E) Implications

- Studying social causes
 - social epi position
 - causal inference position
 - SSNAK position
- Data imagination
 - New sources of old data
 - New sources of new data
 - Managing complex data
- Data use
 - graph database simplify storage and retrieval
 - tidygraph simplifies wrangling
 - MPX NYC augments tidygraph to allow:
 - o grouping nodes
 - o calculating values depending on neighbors
 - o graph descriptors